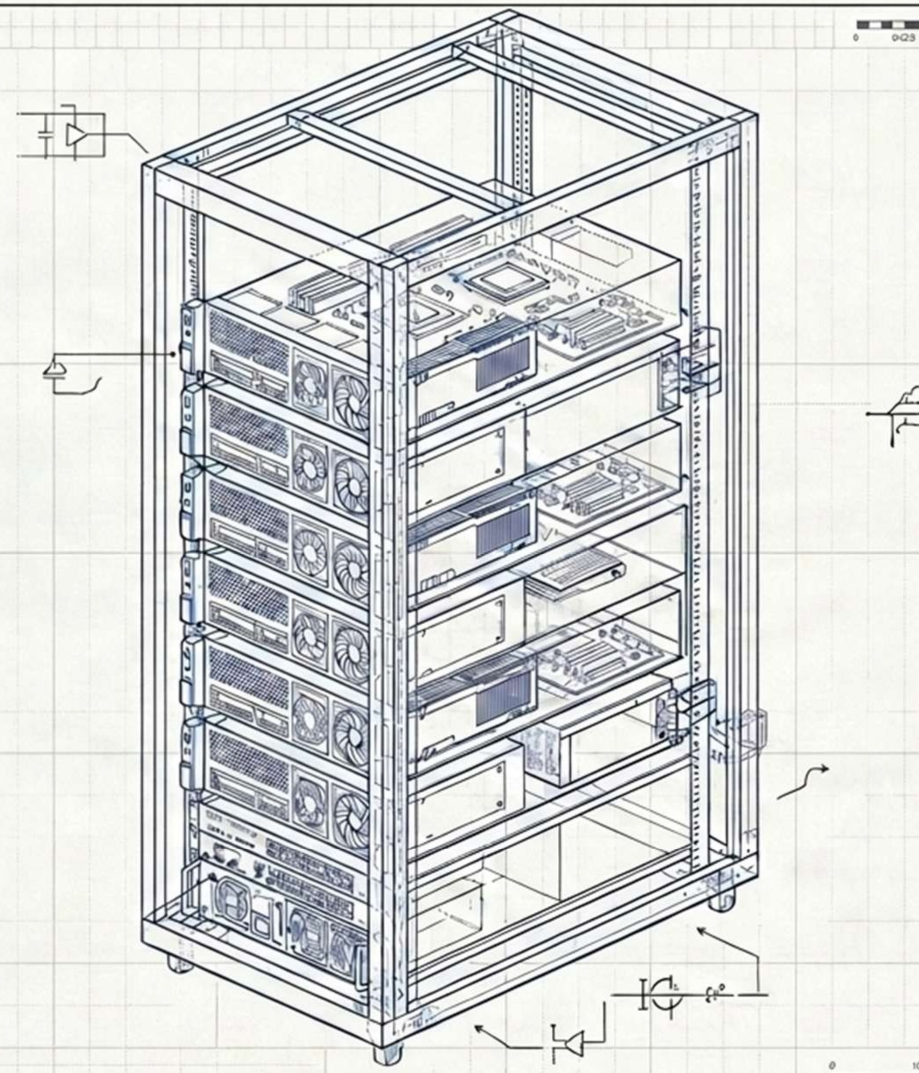
Using Deep Learning to Optimize Energy Consumption in Data Centers and Supercomputers

Author: Hassan Hankir



Content

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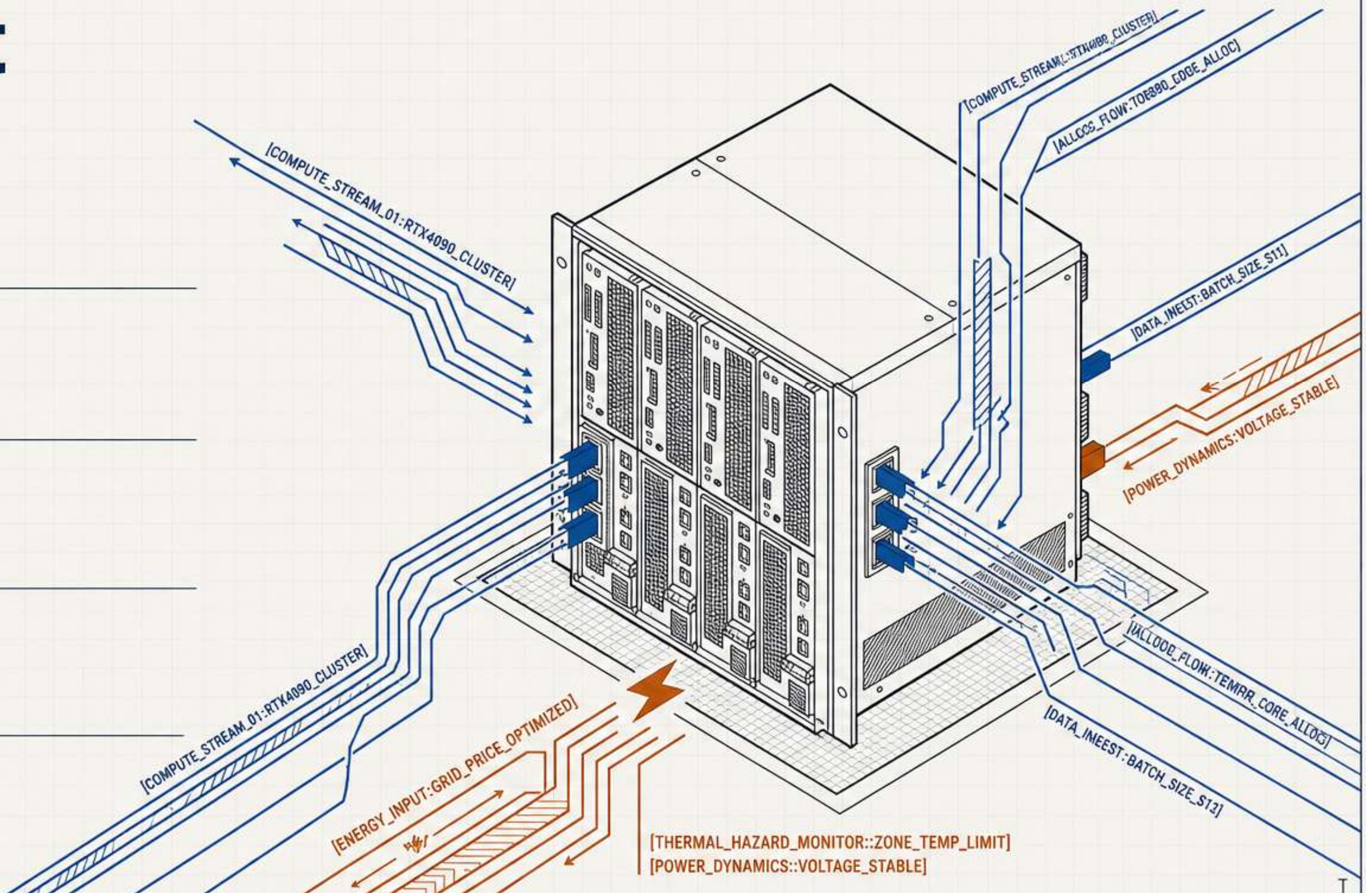
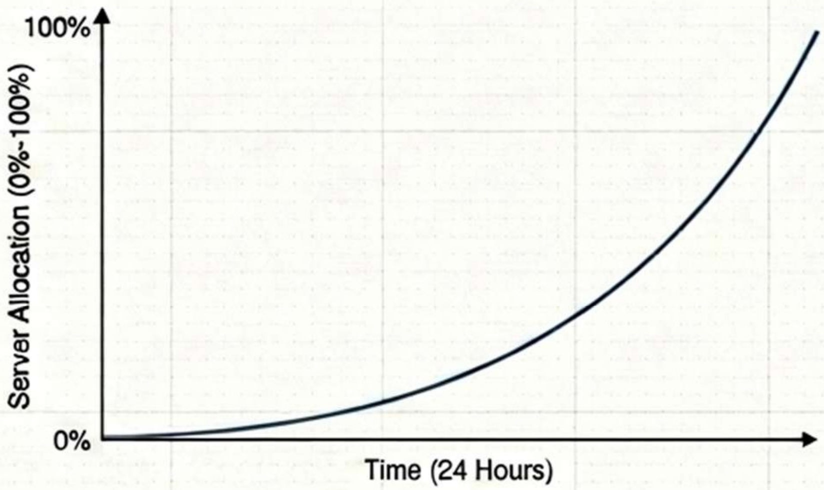
- Solution
- Results
- AI models
- Future work



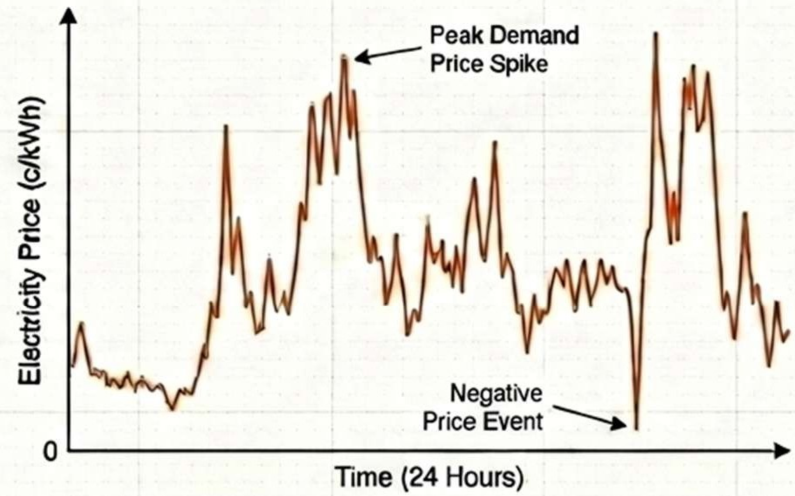
The Problem

The Compute (Static Rules)



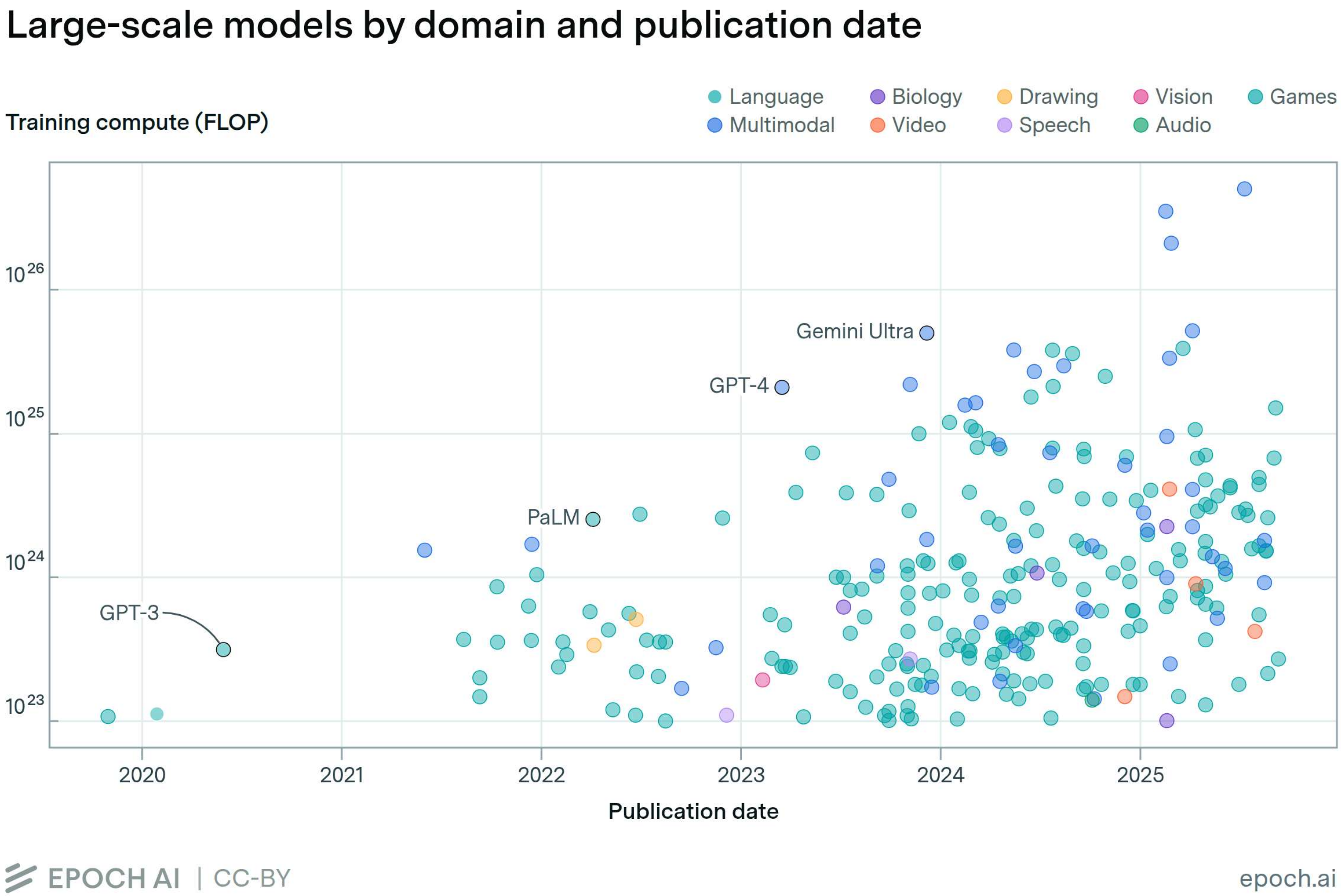
Myopic systems dispatch jobs without anticipating future energy price fluctuations cr complex server dynamics like sleep/wake penalties.

The Environment (Volatile Reality)



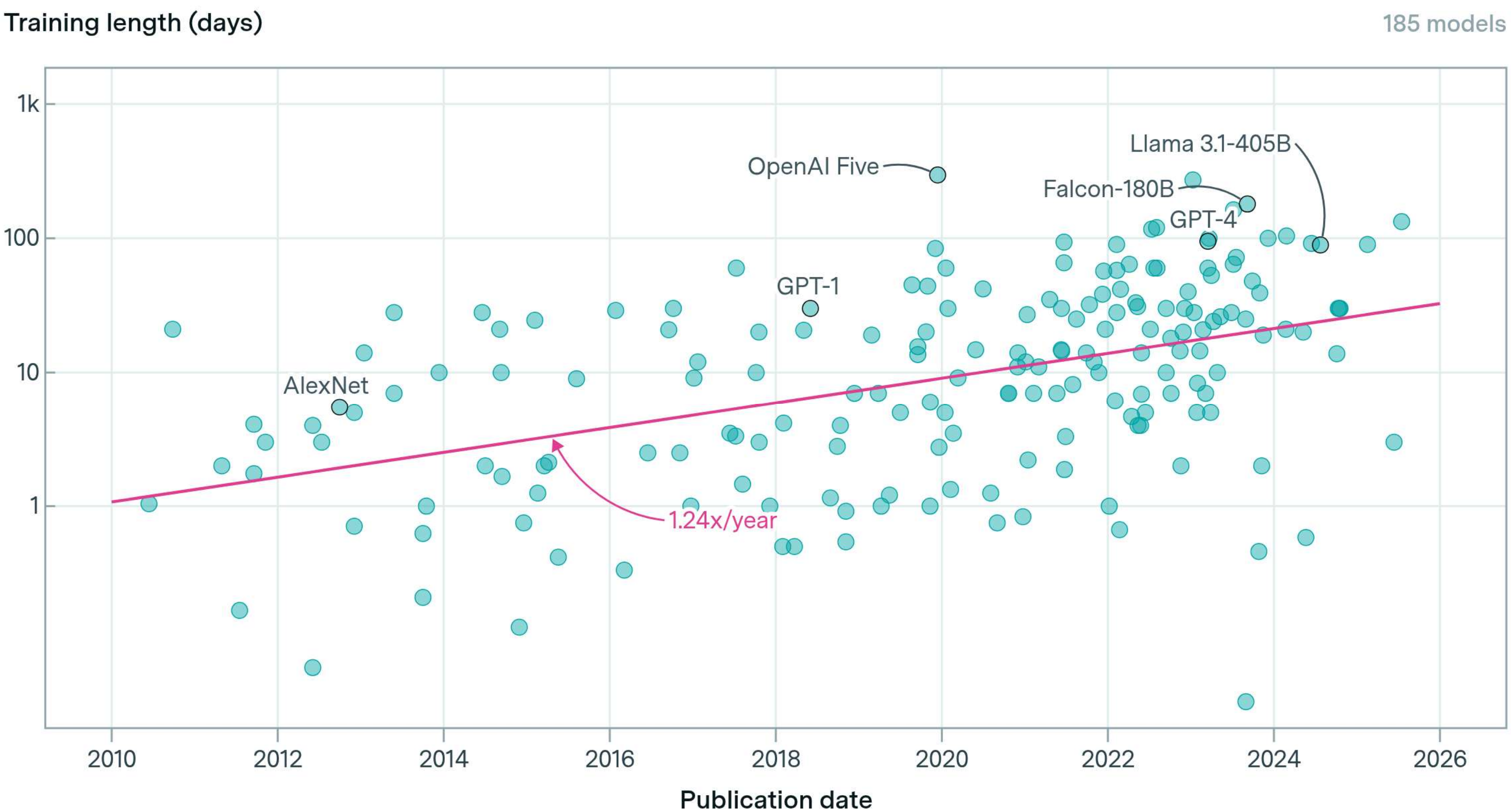
Electricity prices fluctuate due to several factors, including the intermittent availability of renewable energy sources like solar and wind, and significant variations in overall electricity demand throughout the day.

1000s of GPUs and TPUs



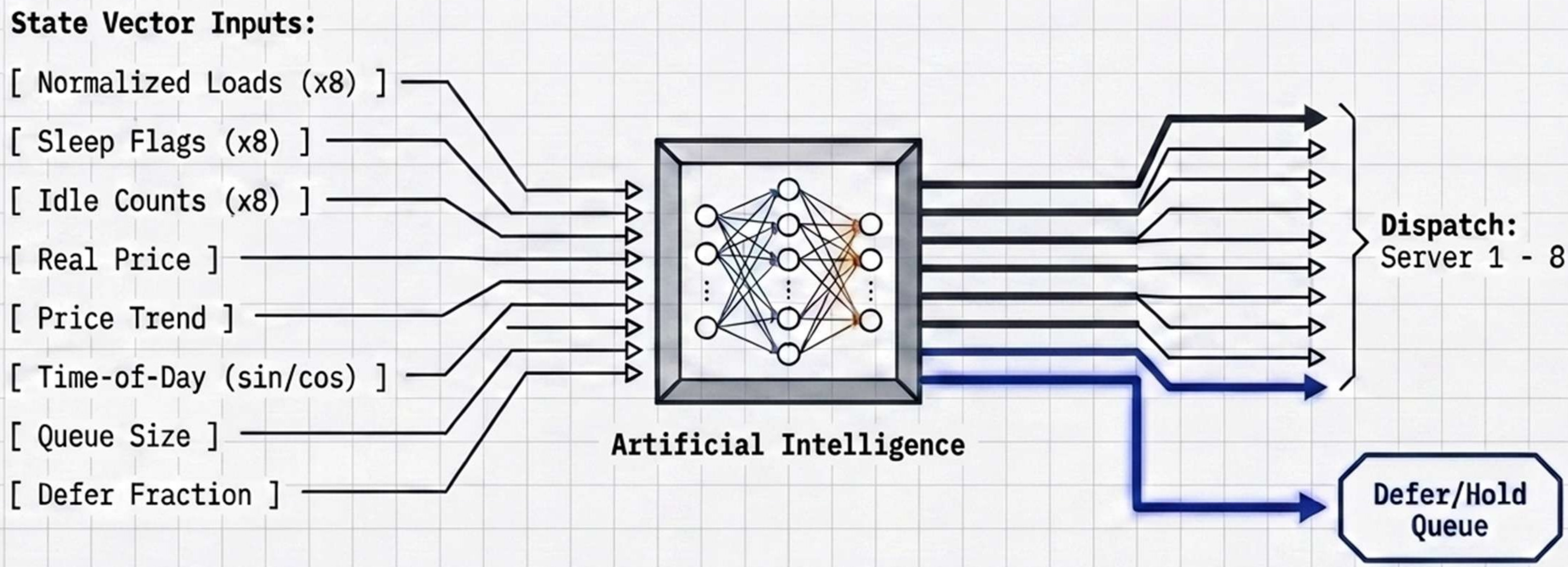
Training length of notable models

Months of training

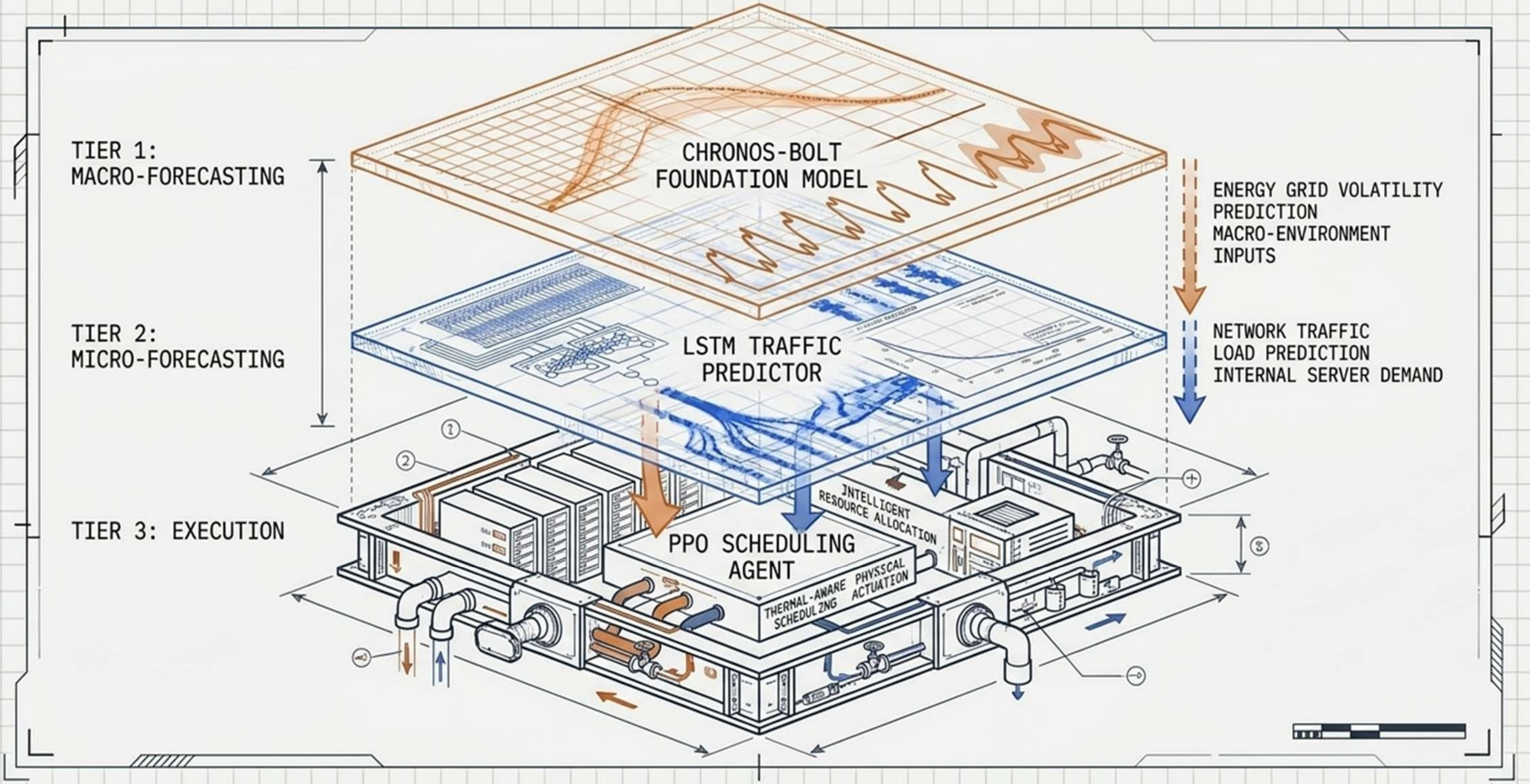


epoch.ai

How can we use AI to reduce energy costs in GPU data center scheduling?



The Autonomous Data Center Stack



Reinforcement learning significantly outperforms heuristic scheduling in dynamic energy markets

The Bottom Line

31.2% **Average Cost Reduction** vs. Energy-Aware Ready (EAR)

33.5% **Average Cost Reduction** vs. Least Loaded (LL)

8 Server GPU Cluster Simulated with Real-World Australian Electricity Demand

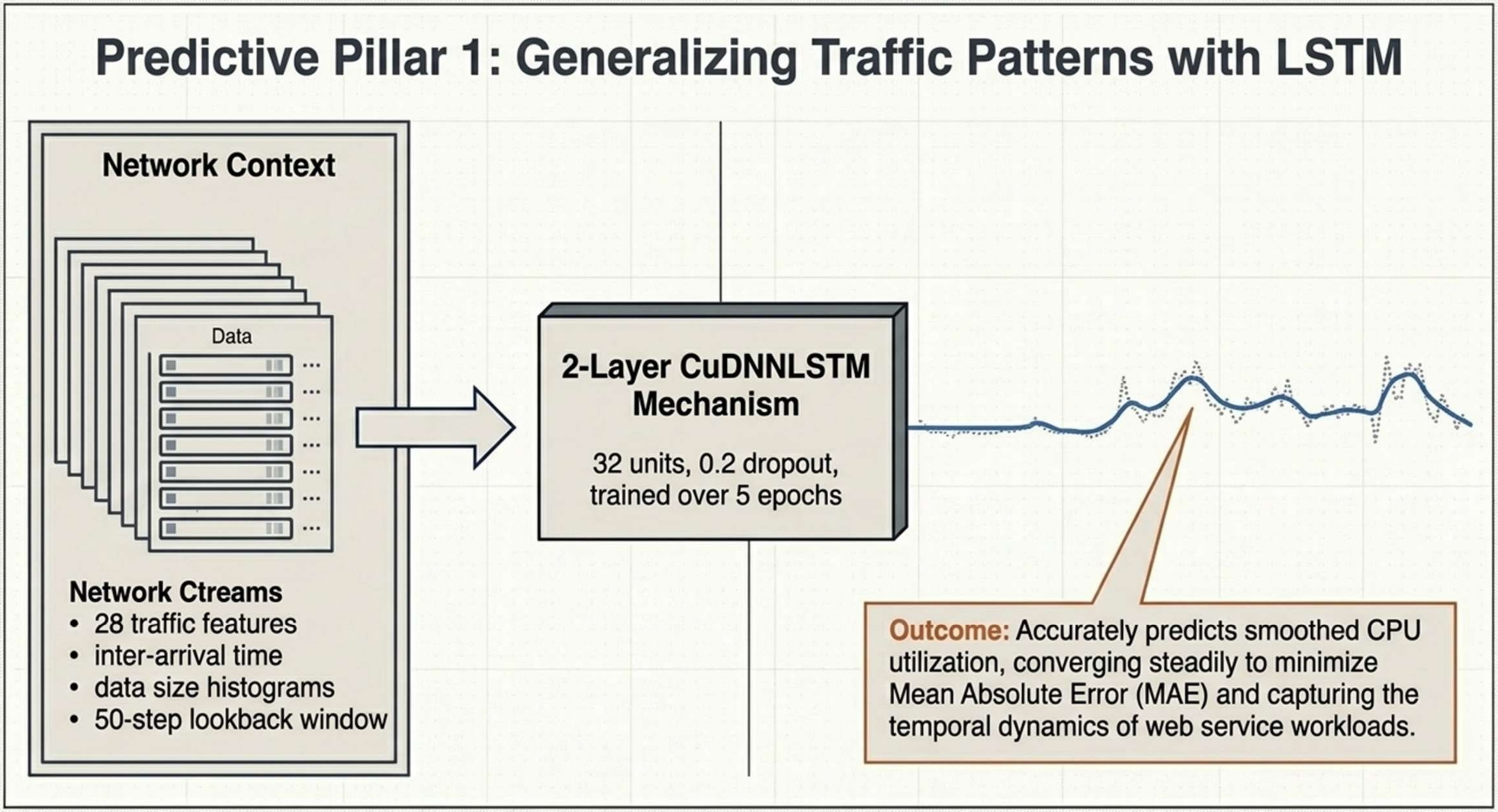
Project Pillars

PPO Agent: Intelligent job dispatch and price-aware deferral

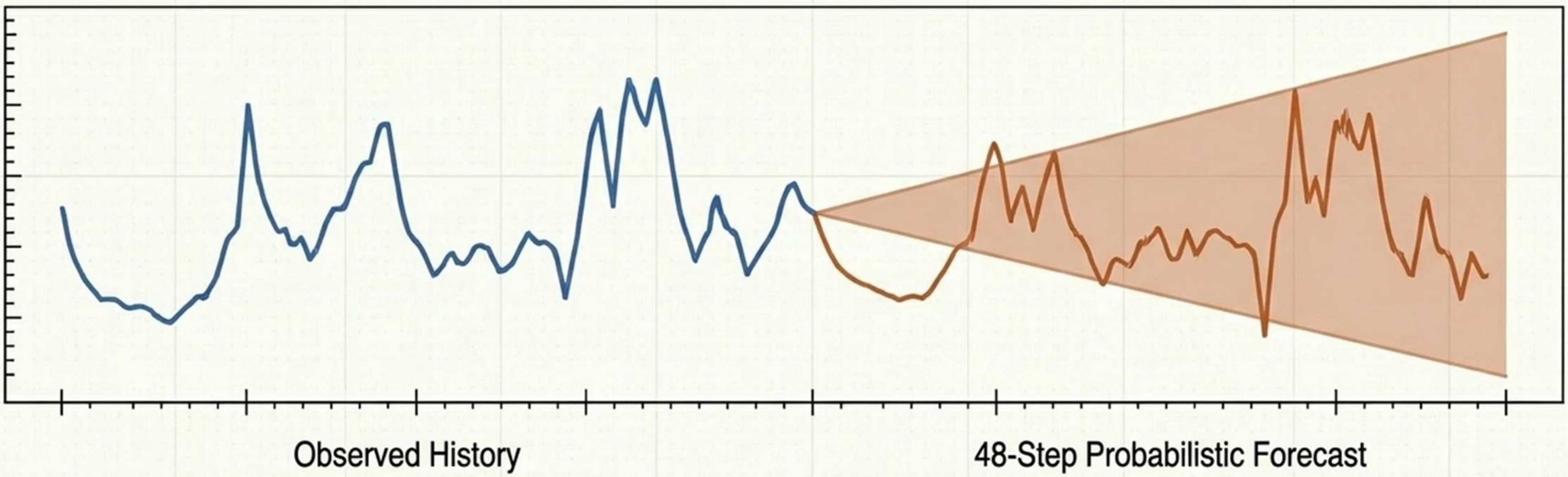
LSTM Forecasting: Multivariate temporal load prediction

Chronos-Bolt: Zero-shot foundational modeling for energy demand

Supported by Ecole Polytechnique, UC Berkeley, and Al Akhawayn University. Special thanks to family, peers, and supervisors.


Predictive Pillar 1: Generalizing Traffic Patterns with LSTM
Network Context
Data
Network Ctreams
• 28 traffic features
• inter-arrival time
• data size histograms
• 50-step lookback window
2-Layer CuDNNLSTM Mechanism
32 units, 0.2 dropout, trained over 5 epochs
Outcome: Accurately predicts smoothed CPU utilization, converging steadily to minimize Mean Absolute Error (MAE) and capturing the temporal dynamics of web service workloads.

Predictive Pillar 2: Zero-Shot Energy Forecasting

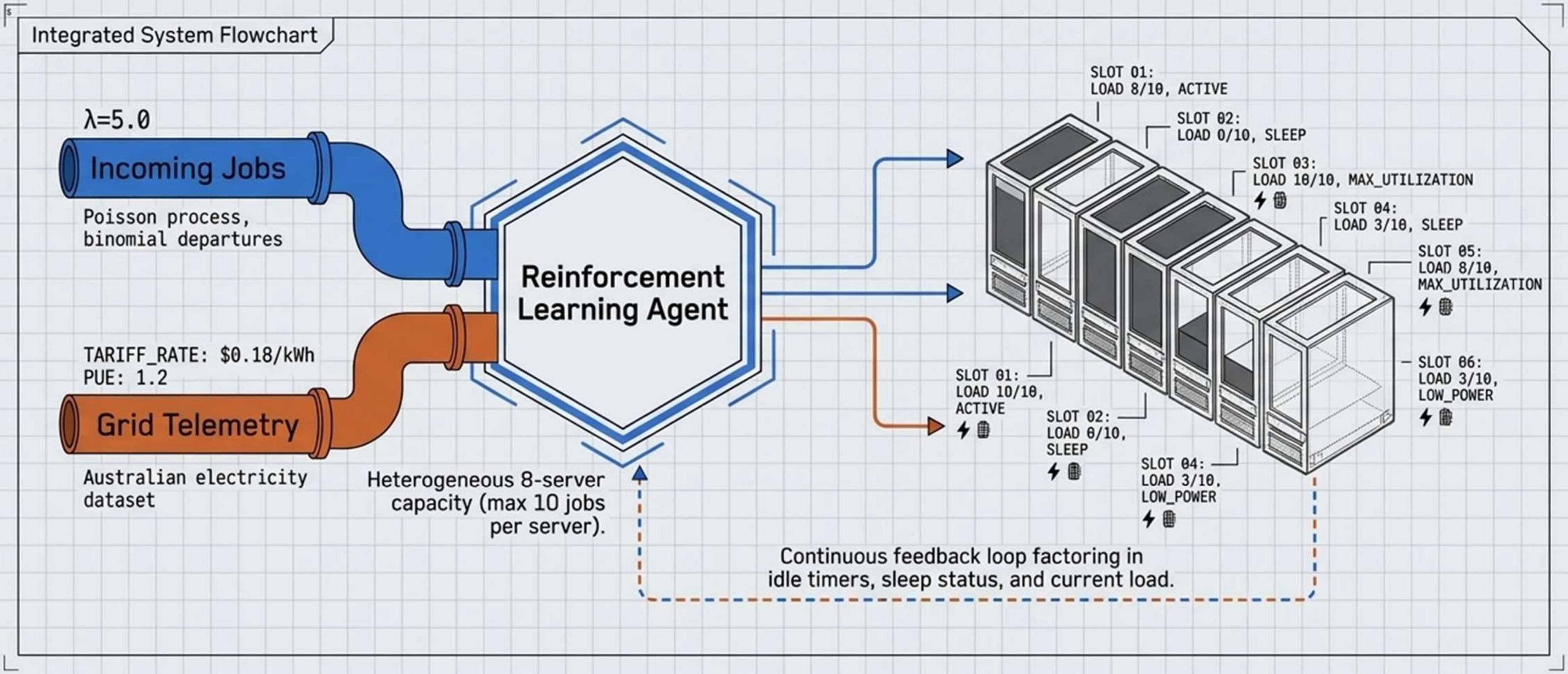


The Model
Amazon's Chronos-Bolt (Small, 48M parameters) via AutoGluon.

The Zero-Shot Advantage
Generates highly accurate 48-step-ahead forecasts on the Australian Electricity subset without any domain-specific fine-tuning.

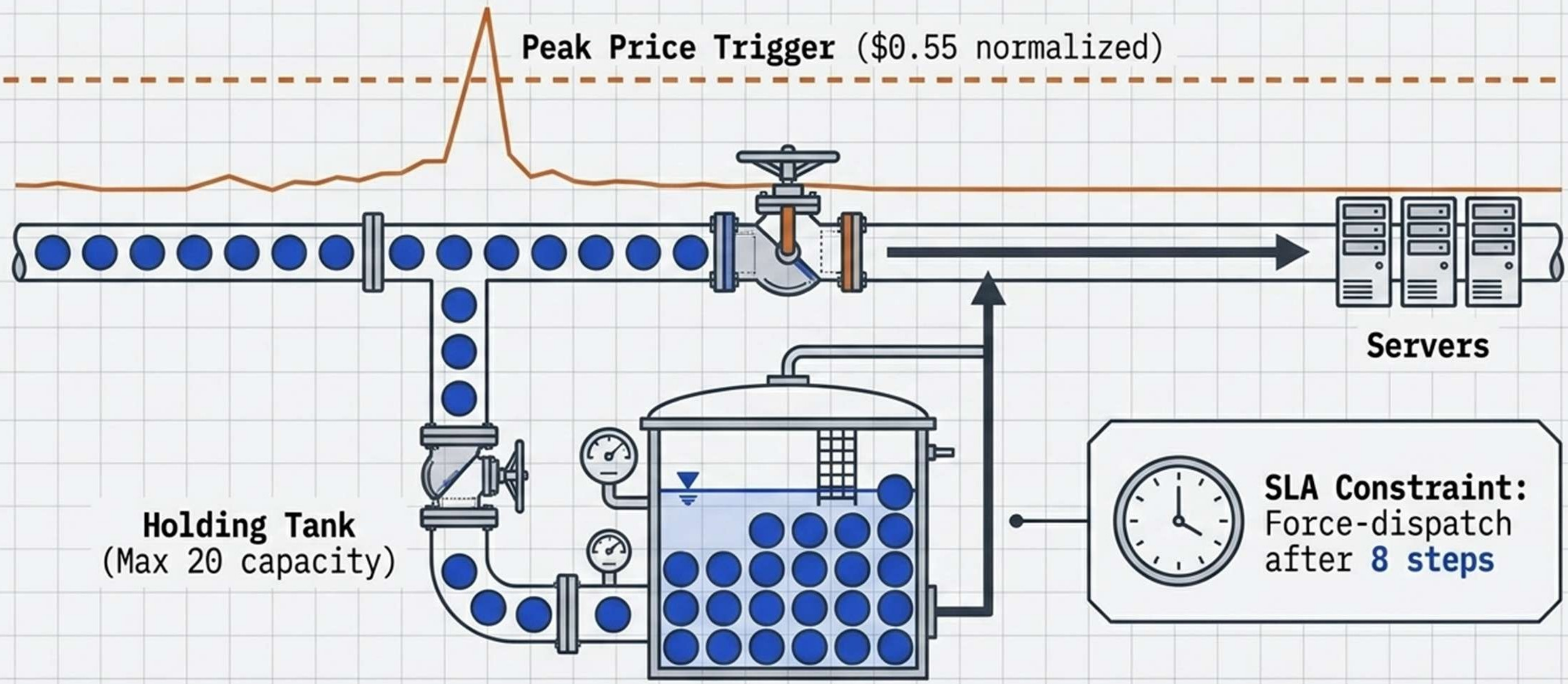
Outcome
Successfully captures recurring daily and weekly seasonality, providing the adaptive scheduling system with calibrated uncertainty bounds for upcoming energy costs.

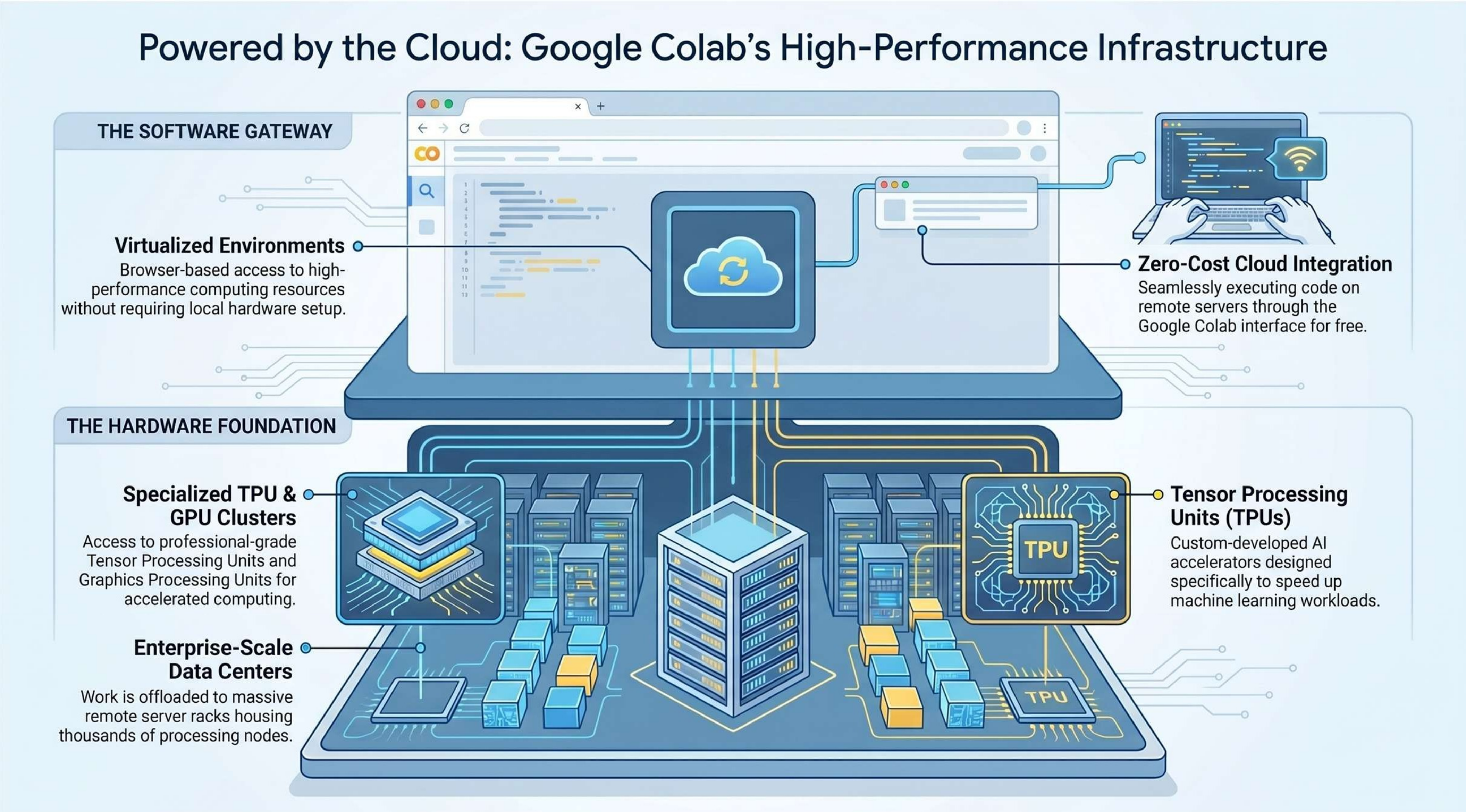
Simulating a dynamic 8-server GPU environment powered by real grid telemetry.



Visual proof of learned price-reactive behavior.

The task deferral mechanism physically buffers workloads to dodge peak pricing.




Powered by the Cloud: Google Colab's High-Performance Infrastructure
THE SOFTWARE GATEWAY
Virtualized Environments
Browser-based access to high-performance computing resources without requiring local hardware setup.
Zero-Cost Cloud Integration
Seamlessly executing code on remote servers through the Google Colab interface for free.
THE HARDWARE FOUNDATION
Specialized TPU & GPU Clusters
Access to professional-grade Tensor Processing Units and Graphics Processing Units for accelerated computing.
Enterprise-Scale Data Centers
Work is offloaded to massive remote server racks housing thousands of processing nodes.
Tensor Processing Units (TPUs)
Custom-developed AI accelerators designed specifically to speed up machine learning workloads.
TPU
TPU

Experimental results

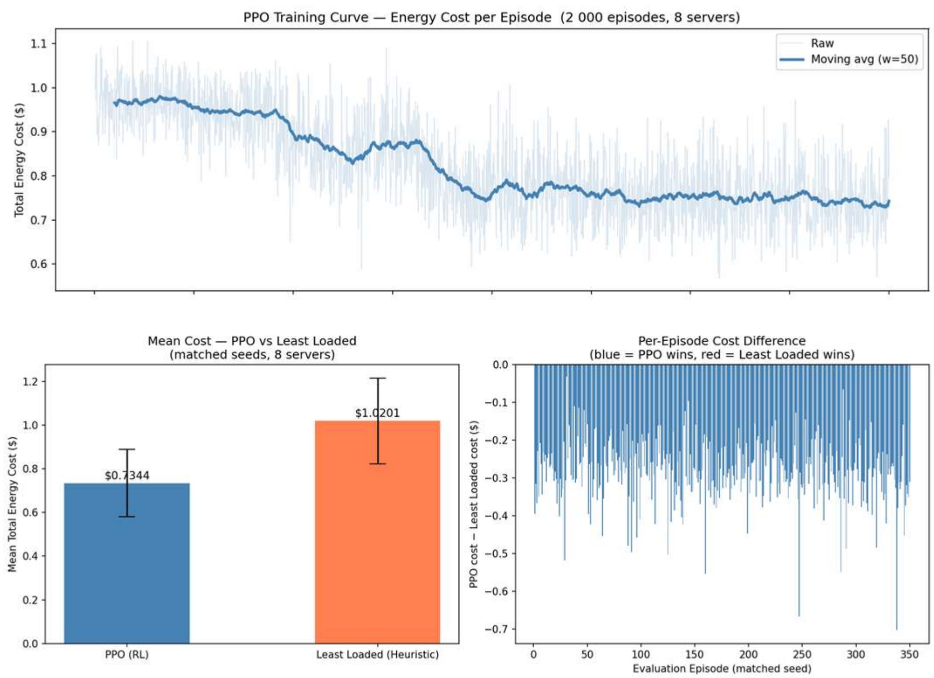
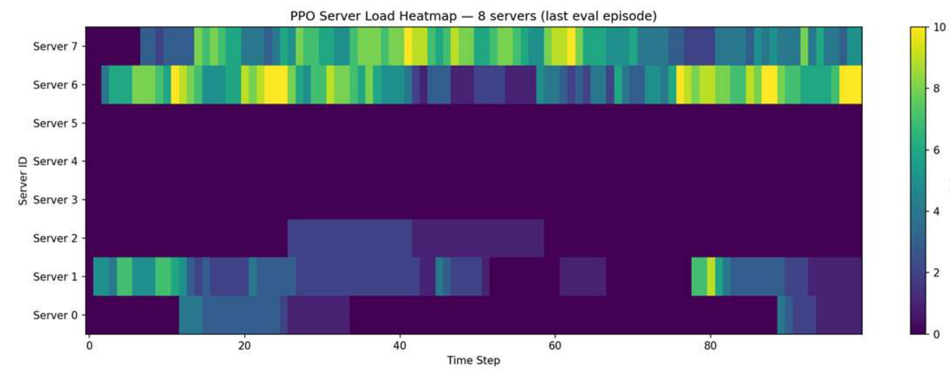
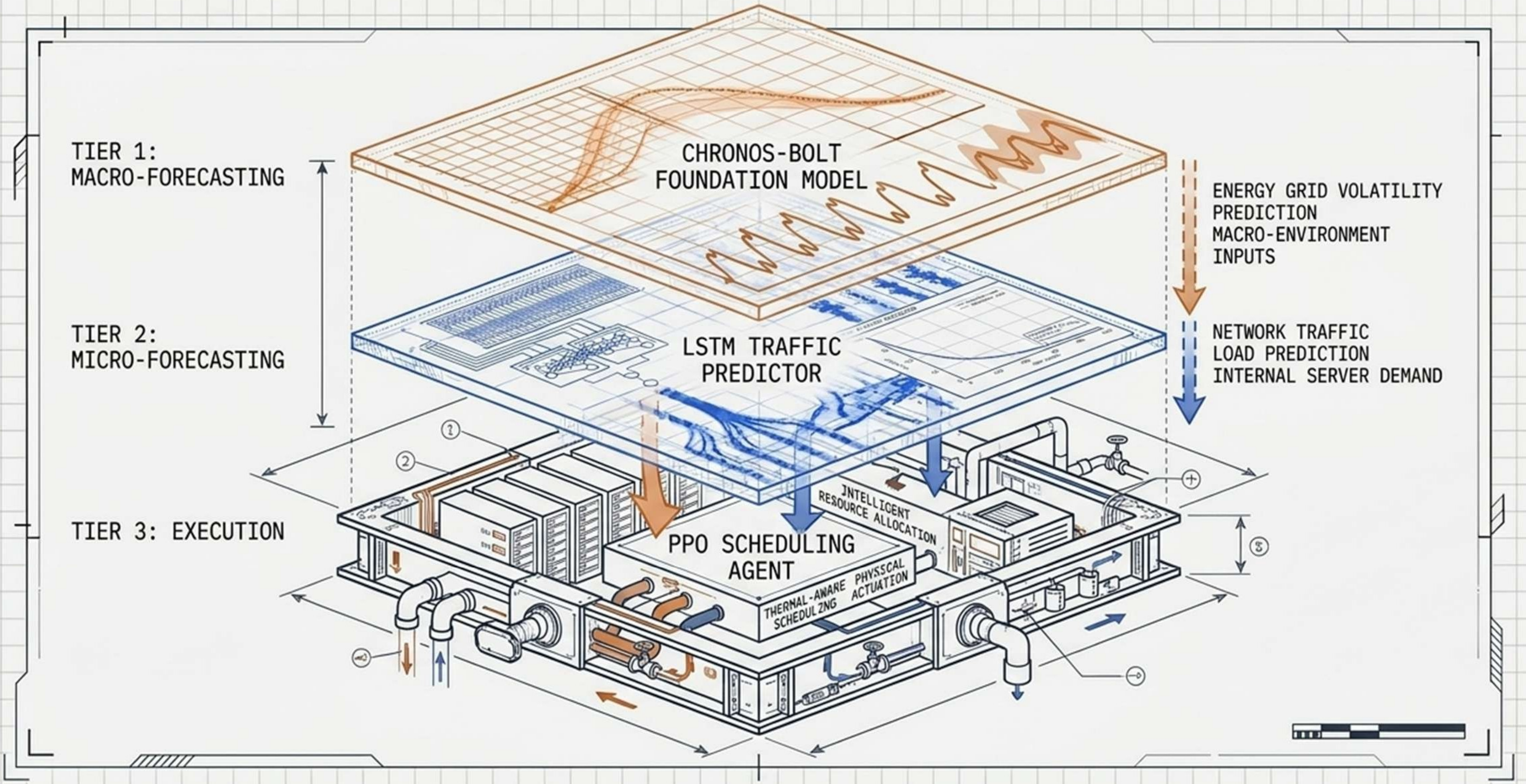


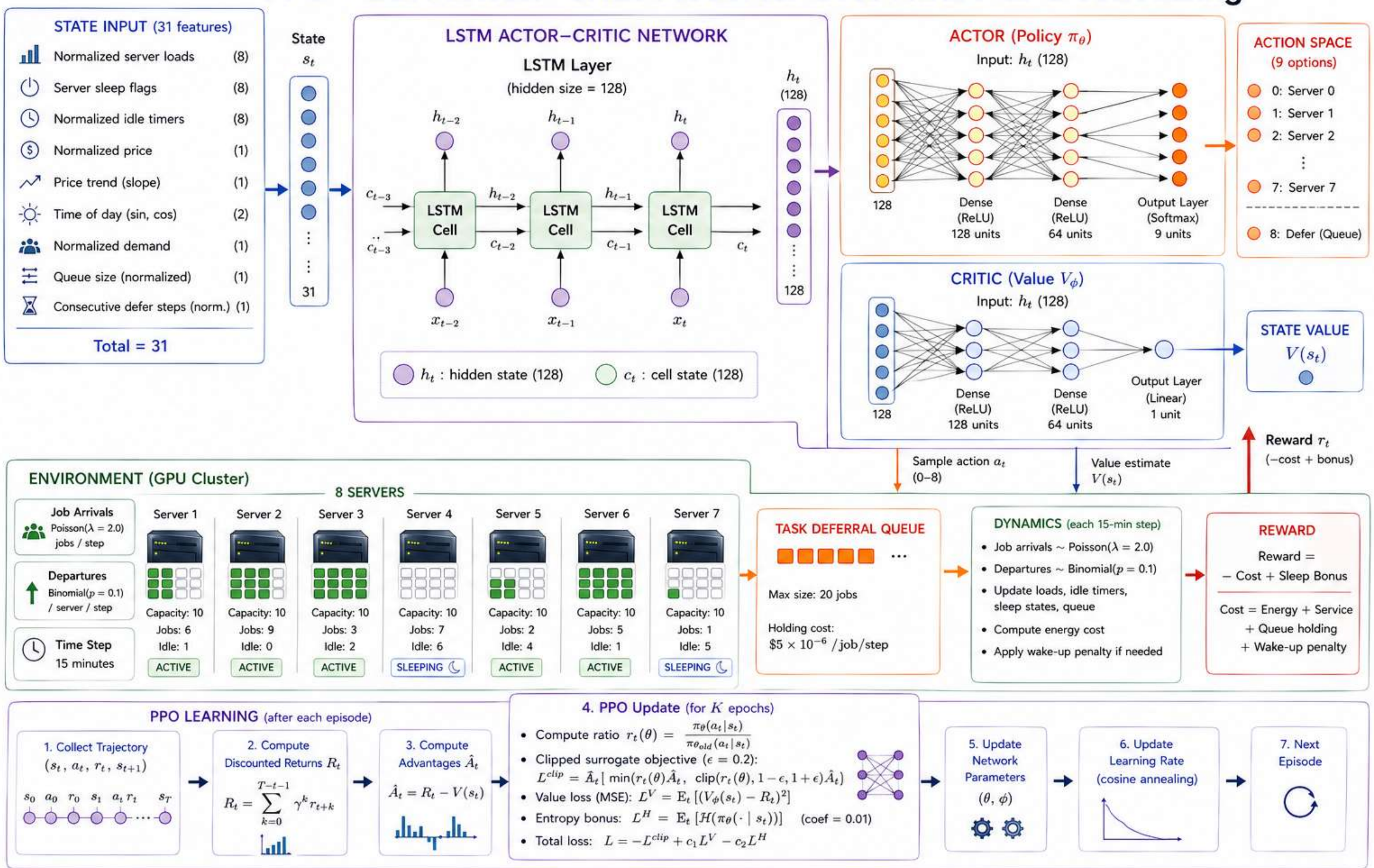
Figure 12. Mean total energy cost comparison between PPO (blue, $0.7344) and Least Loaded (orange, $1.0201) averaged across 5 seeds and 350 evaluation episodes each (left). Error bars represent one standard deviation across seeds. The per-episode cost difference plot (right) shows the distribution of PPO minus Least Loaded cost per matched evaluation episode: the preponderance of negative (blue) bars confirms that PPO wins on a large majority of individual episodes.



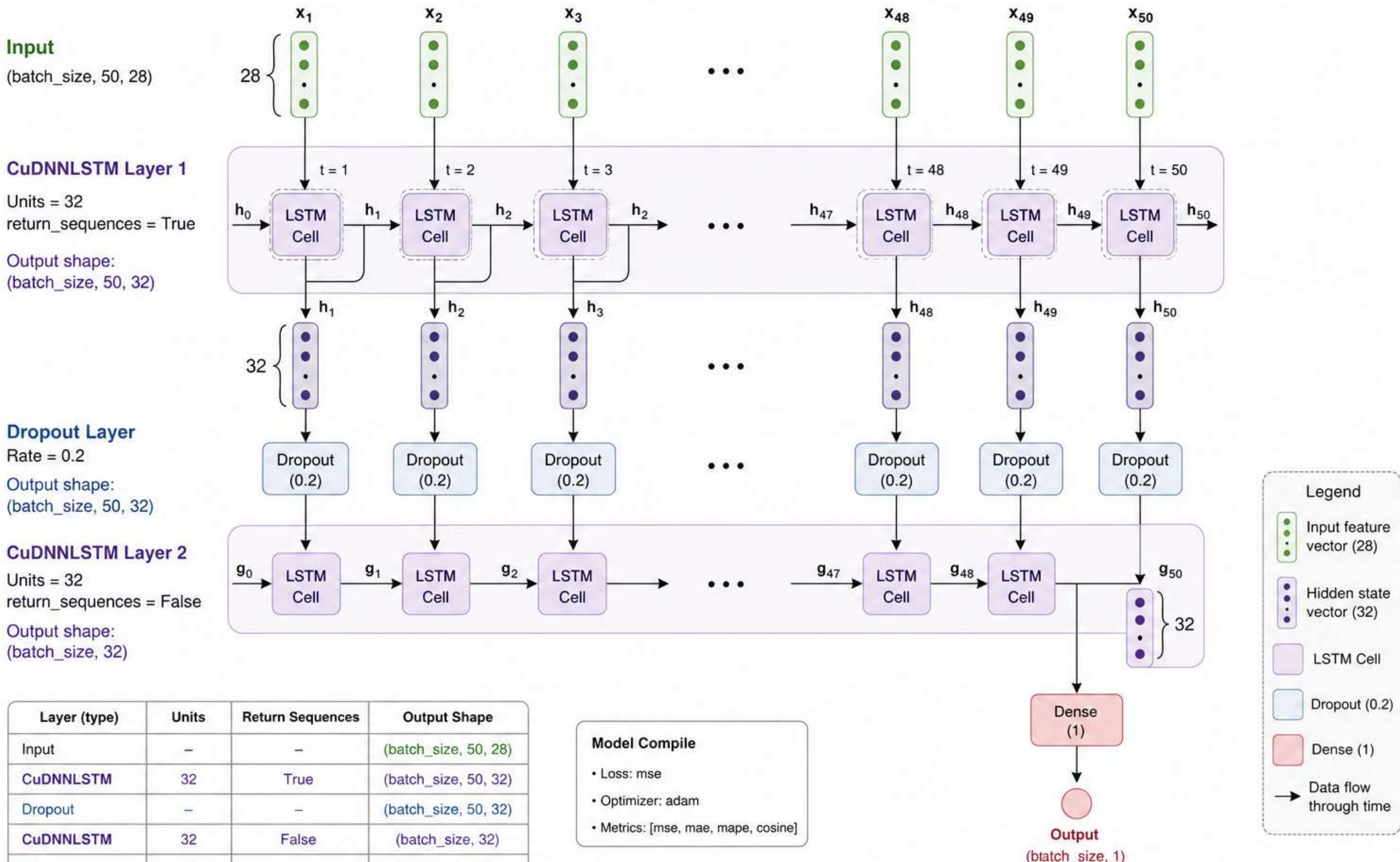
The Autonomous Data Center Stack



PPO + LSTM Actor–Critic Model for Cost-Aware GPU Scheduling



Model Architecture



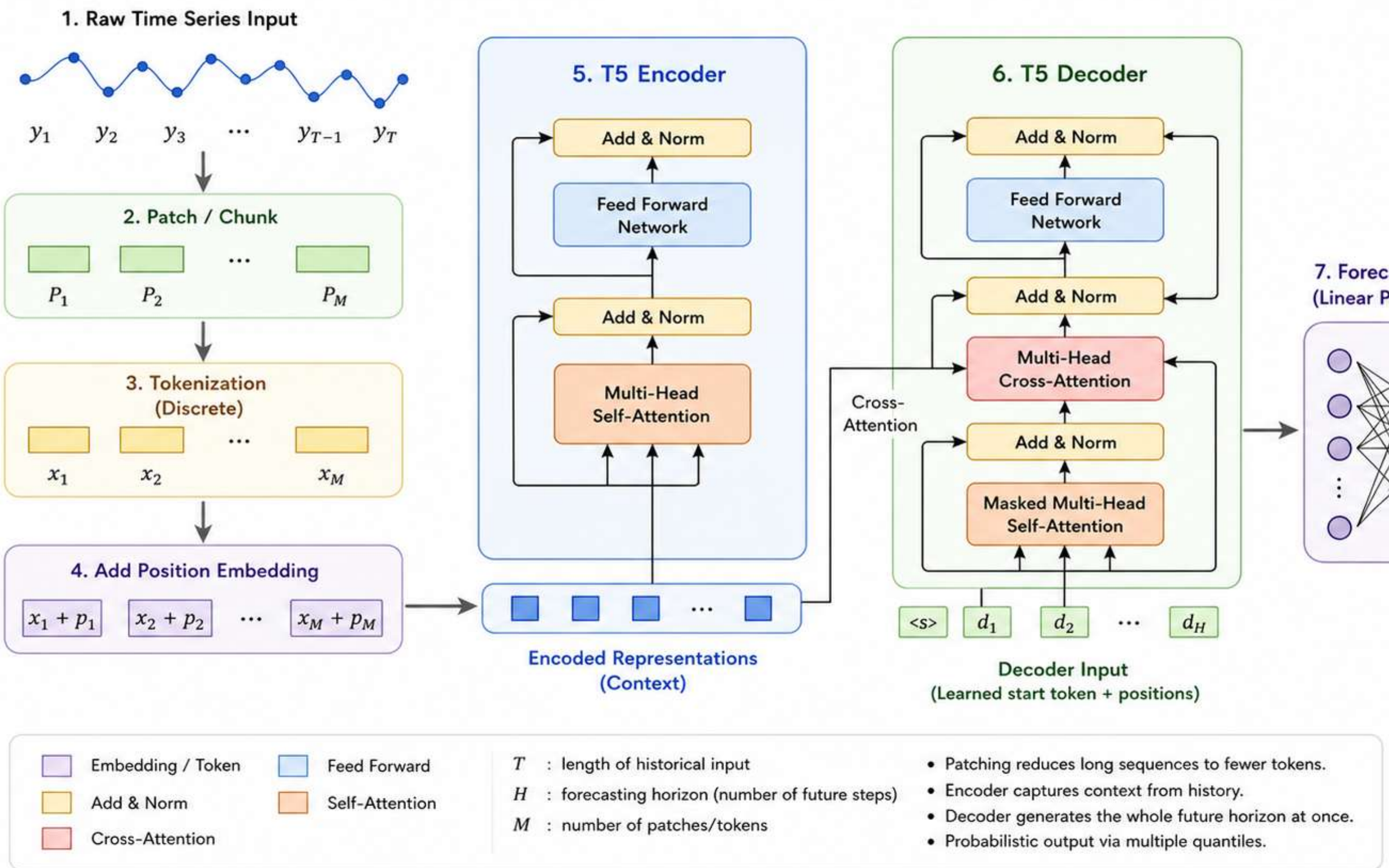
Layer (type)	Units	Return Sequences	Output Shape
Input	–	–	(batch_size, 50, 28)
CuDNNLSTM	32	True	(batch_size, 50, 32)
Dropout	–	–	(batch_size, 50, 32)
CuDNNLSTM	32	False	(batch_size, 32)
Dense	1	–	(batch_size, 1)

Model Compile

- Loss: mse
- Optimizer: adam
- Metrics: [mse, mae, mape, cosine]

Chronos-Bolt Small Architecture (Simplified)

T5 Encoder–Decoder for Time Series Forecasting



Technical Architecture & Reproducibility Hub

Codebase Summary

4 Core Notebooks. Includes LSTM supervised formatting, Chronos AutoGluon pipelines, and dual RL simulation environments.

Academic Output

Conference Proposal Prepared. Full multi-seed protocol and deferral analysis slated for academic submission.

Core References

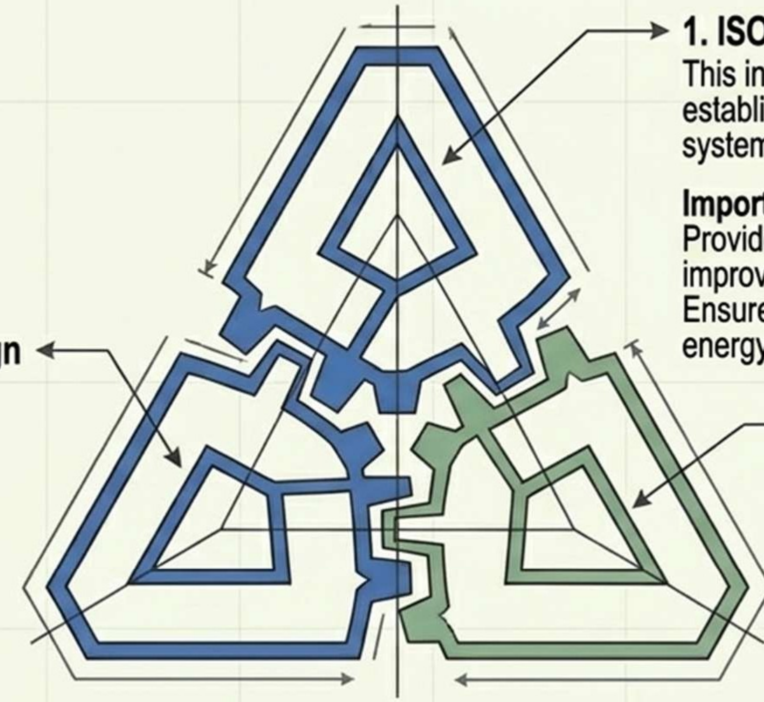
Built on foundational work by Mnih et al. (DQN), Schulman et al. (PPO), and Amazon Research (Chronos).

Repository Link

github.com/hasshank/Deep-Learning-to-optimize-energy-in-data-centers

Project Standards Specification

Aligning deep learning optimization with global engineering and ethical mandates.



1. ISO 50001 – Energy Management Systems.

This international standard specifies requirements for establishing and improving an energy management system.

Importance:
Provides a structured framework for measuring and improving energy efficiency.
Ensures AI optimization aligns with standardized energy management practices.

2. IEEE 2413 – Architectural Framework for the IoT.

Although primarily IoT-focused, this standard addresses large-scale interconnected systems.

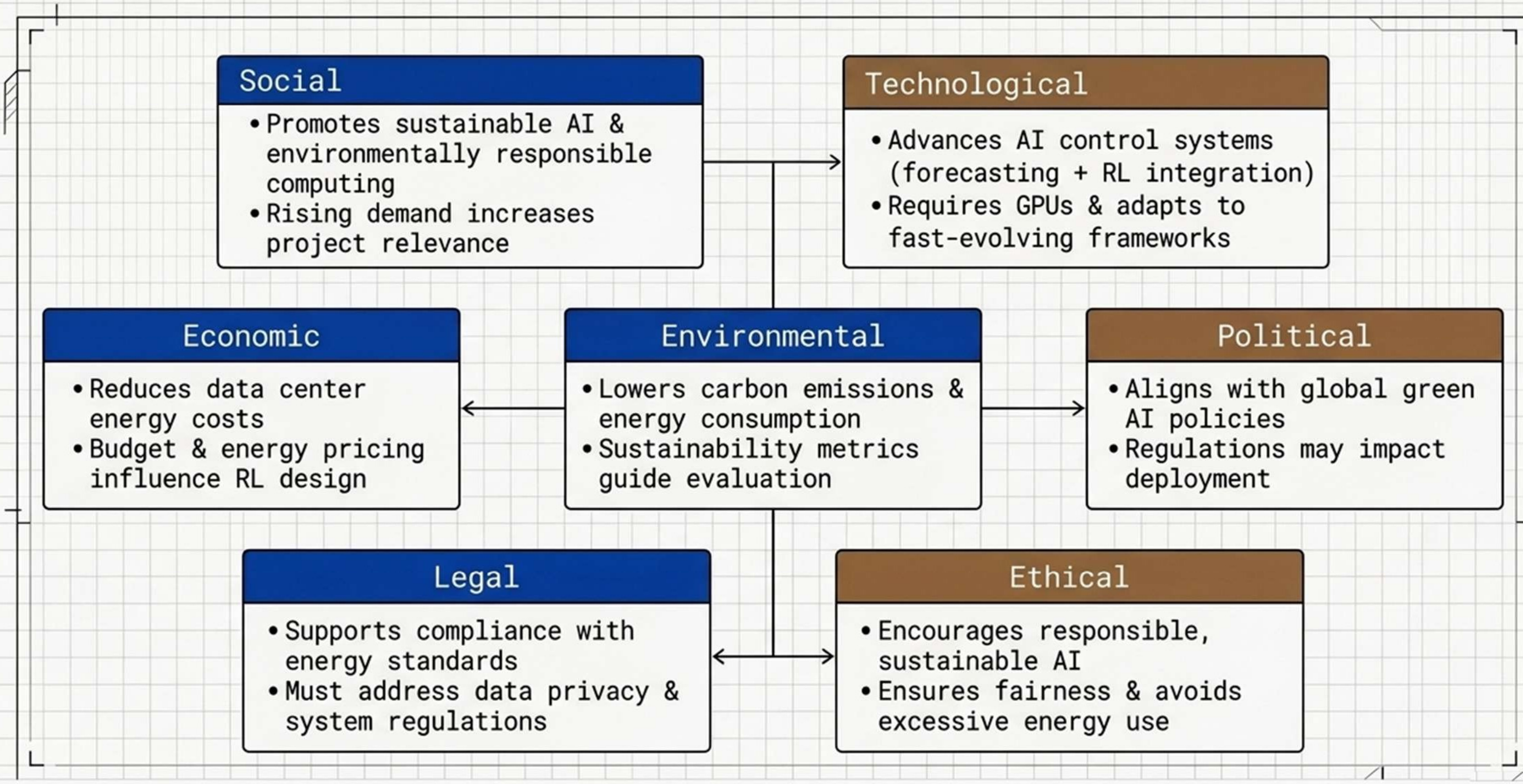
Importance:
Relevant for distributed computing infrastructures. Ensures interoperability and structured system design.

3. IEEE 7000 – Ethical System Design

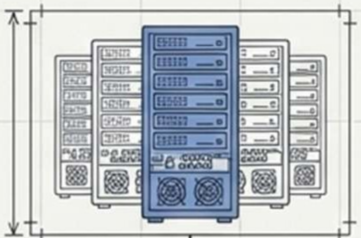
Model process for addressing ethical concerns during system design.

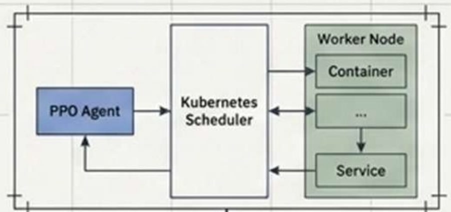
Importance:
Ensures ethical considerations (sustainability, fairness, transparency) are incorporated into AI system design.
Supports responsible AI deployment.

STEEPLE Analysis



Future Work: Scaling to Production





Phase 1: Cluster Expansion

Scaling the environment topology from 8 to 16 and 32 heterogeneous servers to evaluate PPO generalizability and routing latency.

Phase 2: Multi-Objective Carbon Tracking

Upgrading the reward function to incorporate real-time grid Carbon Intensity data alongside electricity price, optimizing for dual sustainability metrics.

Phase 3: Real-Time Deployment

Migrating the Python simulation into a containerized microservice architecture, deploying the PPO agent as a lightweight, real-time Kubernetes scheduler.

Thank you!